DevOps Engineer

Take-Home Assignment

CANDIDATE EXERCISE

Build a simple HTTP web service in a programming language of your choice.

FUNCTIONAL REQUIREMENTS

- Run the server on a port specified by the PORT environment variable at runtime.
- Expose a POST /mean endpoint. Accept a JSON array of numbers in the request body and return the mean.
- Expose a POST /stddev endpoint. Accept the same input and return the population standard deviation.
- Round numeric results to three decimal places. Trailing zeroes do not need to be displayed.

CONTAINERIZATION

Provide a Dockerfile that builds the service into a container image. Also provide one shell script, written in Bash or PowerShell, that builds the image and runs it as a container.

EXAMPLES

```
curl http://localhost:3000/mean -X POST -d '[1,2,3,4,5]' -H "Content-Type: application/json"
3
curl http://localhost:3000/mean -X POST -d '[1.5,3,1,7,4.2]' -H "Content-Type: application/json"
3.34
curl http://localhost:3000/stddev -X POST -d '[1,2,3,4,5]' -H "Content-Type: application/json"
1.414
curl http://localhost:3000/stddev -X POST -d '[1.5,3,1,7,4.2]' -H "Content-Type: application/json"
2.15
```

PROFESSIONAL HANDOFF

Treat your submission as a professional engineering handoff. Organize and document the repository so another engineer can understand, run, review, and maintain it. Some implementation choices are intentionally open-ended; use your judgment and make reasonable decisions.

SUBMISSION AND FOLLOW-UP

- Publish all work, including the service source, Dockerfile, and script, to a public GitHub repository. Send us the repository link within five business days of receiving this assignment.
- **SCREEN SHARING IS REQUIRED FOR THE FOLLOW-UP SESSION.**
  Join from a computer and be prepared to share your screen while running the solution and navigating the repository.
- During the walkthrough, explain your decisions and tradeoffs. Be prepared to answer questions about all submitted work and discuss what you would improve with more time.